

Buffer and Energy-Aware Performance Comparison of Epidemic and Spray & Wait Routing Protocols in Delay Tolerant Networks

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Abstract

Delay Tolerant Networks (DTNs) operate in environments where end-to-end connectivity is intermittent, making efficient routing challenging under limited buffer and energy resources. Epidemic Routing maximises delivery probability through message flooding, but it consumes significant buffer space and energy in an unconstrained environment. In contrast, Spray & Wait restricts replication to reduce resource usage, potentially at the cost of lower delivery rates. This project aims to evaluate and compare the performance of these two protocols under varying buffer and energy constraints to identify the more suitable protocol for resource-constrained DTNs. The study takes on controlled simulations that vary buffer sizes, duty cycles and scan intervals, with performance assessed across three key metrics: delivery ratio, latency, and overhead.

The analysis of experimental data, including fitted performance curves, highlights trade-offs between efficiency and delivery performance. It is expected that Spray & Wait will prove more resource-efficient under strict constraints, whereas Epidemic Routing is expected to perform better when resources are less limited, offering practical insights for DTN protocol selection.

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List of Abbreviations

DTN Delay Tolerant Network
SnW Spray and Wait
ONE Opportunistic Network Environment
TTL Time to live
EA Energy Aware

1 Introduction

1.1 Theoretical Framework and Literature Study

Delay Tolerant Networks (DTNs) are designed to operate in environments where continuous end-to-end connectivity cannot be guaranteed, such as disaster recovery scenarios, space communications, or remote rural areas. Unlike traditional networks, DTNs rely on a store-carry-and-forward mechanism to operate in environments with poor connectivity, where nodes temporarily buffer data and forward it when an opportunity arises [1] [3].

Two well-known DTN routing protocols are Epidemic Routing and Spray & Wait Routing. Epidemic Routing floods messages to all encountered nodes, ensuring high delivery probability but at the cost of high buffer usage and energy consumption [1][2][3]. Spray & Wait Routing, on the other hand, limits the number of message copies by "spraying" a fixed number of copies into the network and letting them "wait" until they reach the destination, thereby reducing resource usage but potentially lowering delivery rates.

In DTNs, energy and buffer constraints play a critical role in determining protocol performance. Limited buffer capacity can lead to message drops due to overflow, while energy constraints can shorten node lifetimes, impacting message delivery rates and delays [1]. Understanding the trade-offs between routing strategies under these constraints is essential for designing efficient DTNs

In our study, we focus on two DTN routing protocols, Epidemic and Spray & Wait. Epidemic routing floods messages to every encountered node, achieving high delivery probability and low latency, but at the cost of excessive buffer use and rapid energy drain.

In contrast, Spray & Wait distributes a fixed number of message copies, thereby reducing resource consumption, but often at the expense of delivery success in low-density networks. Comparative studies by Bylykbashi et al. show that Spray & Wait is more resource-efficient, while Epidemic excels only in resource-rich conditions. Prior work has generally considered either buffer constraints or energy limitations, but rarely in combination. Our project addresses this gap by conducting a simulation-based comparison of Epidemic and Spray & Wait under varying buffer and energy levels.

By quantifying trade-offs in delivery ratio, latency, and energy consumption, we aim to provide insights into which protocol is more suitable under resource-constrained environments

1.2 Research Questions and Hypothesis

This study is guided by the hypothesis that the Spray & Wait routing protocol will be more resource-efficient under buffer and energy constraints, whereas the Epidemic routing protocol will achieve higher delivery ratios when such resources are less constrained.

To investigate this, the research will address two central questions:

- i. What are the performance trade-offs between Epidemic and Spray & Wait routing protocols under buffer and energy constraints in Delay Tolerant Networks (DTNs)?
- ii. How do these constraints influence key performance metrics such as delivery ratio, latency, and energy consumption?

2 Method

2.1 Overview

We conducted a quantitative, simulation-based study in The ONE simulator to compare two DTN forwarding strategies, i.e., Epidemic and SnW, under varying per-node buffer sizes, duty cycle and scan intervals. We modelled pedestrians and cars on a map-based Helsinki layout as shown in Figure 1. All runs used the same mobility, traffic, and radio parameters, so differences arise only from the routing protocol and the buffer size.

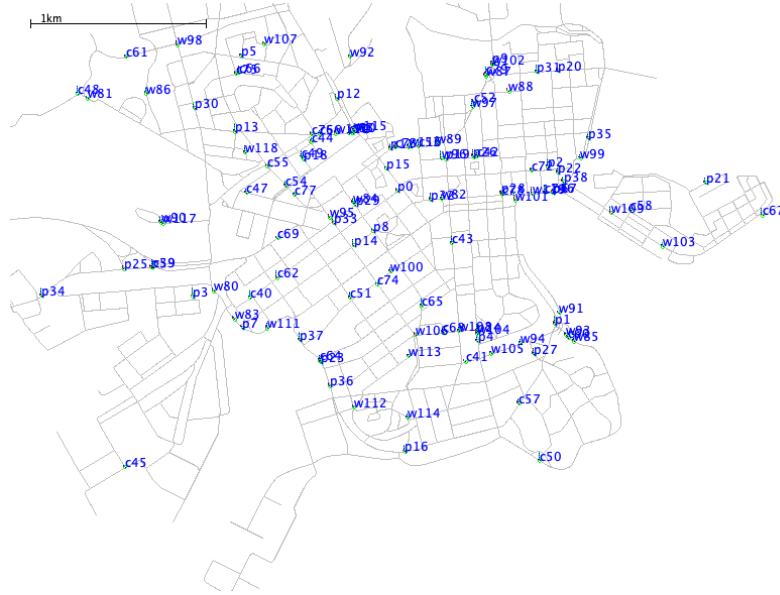


Figure 1 : Simulation environment.

2.2 Simulator Overview

All experiments were conducted using **The Opportunistic Network Environment (The ONE)** simulator, chosen because it includes built-in implementations of Epidemic and Spray & Wait routing protocols, as well as their energy-aware versions. This allowed us to focus on analysing performance outcomes rather than implementing routing algorithms.

The ONE supports realistic DTN modelling with configurable parameters such as buffer size, duty cycle, and scan interval. In our study, we modified the simulator's setting files to modify these parameters and control experimental conditions.

After each simulation run, the ONE automatically generated detailed log and report files containing metrics such as delivery rate, delay, forwarding overhead, hop count, and aborted transfers. We aggregated these outputs, cleaned the data, and plotted performance curves to compare the behaviour of Epidemic and Spray & Wait under varying buffer and energy constraints.

2.3 Environment

- **Run length:** 12 h simulated time (43200 s).
- **Nodes:** 120 total (3 groups $\times$ 40).

- **Pedestrians (groups p, w):** 0.5–1.5 m/s.
- **Cars (group c):** road-restricted movement, 2.7–13.9 m/s.
- **Map:** Helsinki road & pedestrian layers; world size $\approx 4.5 \text{ km} \times 3.4 \text{ km}$.
- **Interface:** one Bluetooth-like link per node, 10 m range, 250 kB/s throughput.
- **Traffic**
 - **Generator:** one random source–destination event generator.
 - **Inter-arrival:** 25–35 s (uniform).
 - **Message size:** 0.5–1 MB.
 - **TTL:** 300 min (18,000 s)
 - **Per run:** 1,460 messages created (from simulator logs).

2.4 Variables and metrics

- **Independent variables:**
 - **Routing protocol** (Epidemic vs Spray and Wait & their Energy-Aware versions)
 - **Buffer size** (1–50 MB): Per-node storage for messages waiting to be forwarded.
 - **Scan interval** (1, 15 & 30): how often a node looks for neighbours (discovery sweep period). Shorter/frequent scans find more contacts and boost delivery, but increase energy use.
 - **Duty Cycle** (1000;1200 & 2000;2400): the radio's awake/sleep pattern per node (e.g., 1000;1200 = 1.0 s ON, 1.2 s OFF, repeating). Longer ON windows raise the chance of detecting peers and finishing transfers, but use more energy.
- **Dependent metrics:**
 - **Delivery rate (%)**: Percentage of created messages that arrive before TTL.
 - **Delivery delay (s)**: Time from message creation to delivery.
 - **Forwarding overhead ratio**: Extra forwards per delivered message.
 - **Aborted transfers**: Transfers that were initiated but not completed.
 - **Average hop count**: Mean number of hops between source and destination for delivered messages.

2.4.1 Choice of Parameters

- **Buffer size (1–50 MB)**: Covers devices from highly constrained (1 MB) to resource-rich (50 MB), highlighting where delivery rates begin to saturate.
- **Scan interval (1, 15, 30 s)**: Represents aggressive, moderate, and energy-saving neighbour discovery rates to observe delivery energy trade-offs.
- **Duty cycle (1000;1200 ms, 2000;2400 ms)**: Models short and long radio awake periods, reflecting low and medium power operation.

2.5 Procedure

- **For buffer analysis**
 1. Fix mobility, traffic, and radio parameters; warm-up 1,000 s.
 2. Run Epidemic for 12 h; export to reports.
 3. After each simulation, the buffer size is increased.
 4. Run Spray and Wait with the same scenario; export the same reports.
 5. Aggregate results into a single table.

6. Produce three plots: Delivery vs Buffer, Delay vs Buffer, and Overhead vs Buffer.

- **For energy-aware analysis:**

1. Fix mobility, traffic, and radio parameters; warm-up 1,000 s.
2. Run Epidemic for 12 h; export to reports.
3. After each simulation, the duty cycle, buffer size and scan interval are increased.
4. Run Spray and Wait with the same scenario; export the same reports.
5. Aggregate results into their respective tables.
6. Produce plot: Delivery Rate vs Scan Interval for two sets of duty cycle.

2.6 Ethics

No human subjects or personal data were collected. All mobility and traffic are synthetic.

3 Results and Analysis

We evaluated Epidemic and SnW across buffer sizes of 1, 2, 5, 10, 20, and 50 MB. Results are summarised in Table 1.

Routing Protocol	Buffer Size (MB)	Delivery Rate (%)	Average Delivery Delay (s)	Median Delivery Delay (s)	Forwarding Overhead Ratio	Aborted Transfers	Average Hop Count
Epidemic	1	3.77	1848.25	1232.4	144.02	9365	4.58
	2	7.53	2160.07	1593.6	169.41	18280	5.04
	5	19.45	3692.55	2978.9	85.43	22218	4.38
	10	33.63	4907.4	4122.1	52.72	22698	3.54
	20	50.21	6681.09	6077.4	35.23	22858	2.68
	50	63.56	7774.24	7203.5	27.44	22847	2.26
SnW	1	6.16	1055.16	1015.4	39.92	5180	2.06
	2	14.66	1524.88	1390	25.11	8883	2.09
	5	38.01	2604.72	2374	12.41	11828	2.1
	10	62.26	3921.68	3478.1	7.71	12399	2.11
	20	77.26	5264.14	4497.4	6.21	12750	2.13
	50	78.7	5407.73	4627.1	6.09	12793	2.13

Table 1: Summary metrics by protocol and buffer (delivery %, median delay, overhead, aborted, hops).

After, we ran the energy-aware version of each protocol across different Duty Cycles, buffer size and scan intervals. Results are summarised in Table 2 and Table 3.

Duty Cycle (ms)	Buffer Size (MB)	Scan Interval (s)	Delivery Rate (%)	Median Delivery Delay (s)	Forward Overhead Ratio
1000;1200	10	1	0.34	2036.7	75.6
		15	10.41	2715.1	55.5329
		30	19.45	3244.6	54.3592
	50	1	0.34	2036.7	76.4
		15	14.59	3563.9	37.9437
		30	28.9	4850.9	34.9929
2000;2400	10	1	1.23	1498.7	58.1667

		15	22.53	3367.2	53.4438
		30	36.58	3678.1	50.809
	50	1	1.16	1498.7	63.0588
		15	35.48	5423.6	33.1023
		30	66.03	7001.3	27.6712

Table 2: Results for Energy-Aware epidemic protocol (delivery %, median delay, overhead, aborted, hops)

Duty Cycle (ms)	Buffer Size (MB)	Scan Interval (s)	Delivery Rate (%)	Median Delivery Delay (s)	Forward Overhead Ratio
1000;1200	10	1	0.34	2036.7	75
		15	14.38	2643.4	37.8095
		30	26.64	2862.9	37.3033
	50	1	0.34	2036.7	75.8
		15	18.77	3134.1	28.938
		30	39.93	3930.1	24.7136
2000;2400	10	1	1.16	1498.7	60
		15	30.48	2952.1	36.836
		30	51.44	3303.2	33.1944
	50	1	1.16	1498.7	61.0588
		15	47.74	4324.2	23.7403
		30	83.49	4740.7	20.95

Table 3: Results for Energy-Aware Spray and Wait protocol (delivery %, median delay, overhead, aborted, hops).

3.1 Analysis for Buffer

3.1.1 Delivery vs. Buffer

Figure 2 shows that SnW consistently delivers more messages than Epidemic at every buffer size. At 1 MB, delivery is low for both (Epidemic 3.77%, SnW 6.16%), indicating that storage is the dominant bottleneck. As the buffer rises, delivery increases sharply. The knee of the curve lies around 10–20 MB: at 10 MB, SnW reaches 62.26% vs 33.63% for Epidemic; at 20 MB, SnW reaches 77.26% vs 50.21% for Epidemic. Beyond 20 MB, delivery gains are modest (SnW 78.70% at 50 MB; Epidemic 63.56%), indicating diminishing returns.

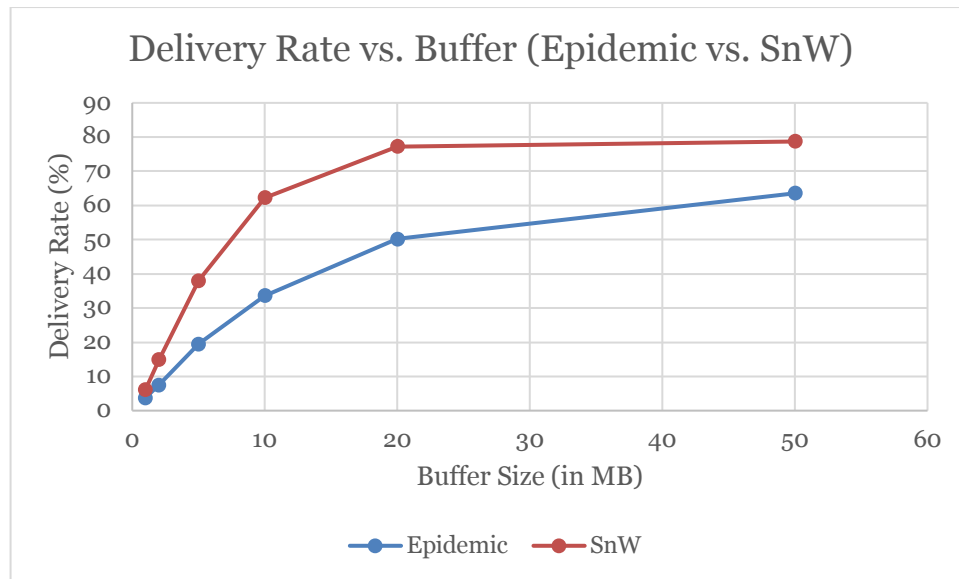


Figure 2 : Delivery rate vs. buffer size (Epidemic vs. Spray-and-Wait).

3.1.2 Delay vs. Buffer

Figure 3 plots the median delay. Both protocols see increasing delay as buffers grow, the message queue gets longer and can take longer paths. At every buffer, Spray and Wait's median delay is lower than Epidemic's. For example, at 10 MB, the medians are 3478 s (SnW) vs 4122 s (Epidemic); at 20 MB, they are 4497 s vs 6077 s.

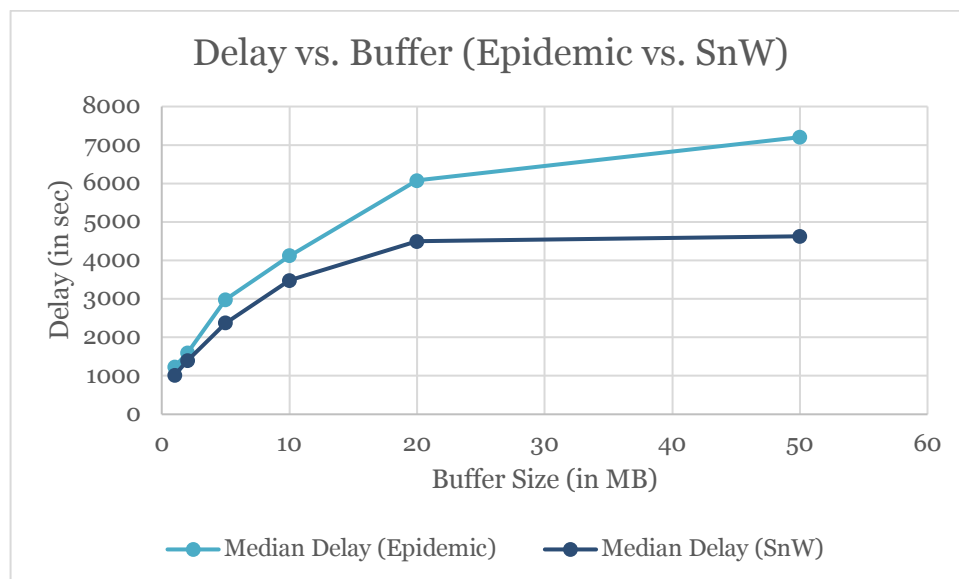


Figure 3: Median delivery delay vs. buffer size (Epidemic vs. Spray-and-Wait).

3.1.3 Forwarding Overhead vs. Buffer

Figure 4 shows the forwarding overhead ratio. Spray and Wait's overhead is an order of magnitude lower across the range (e.g., at 10 MB, 7.71 for SnW vs 52.72 for Epidemic). Overhead declines as buffers increase for both protocols, but the gap remains large. Furthermore, at very small buffers for the epidemic, frequent drops and retransmissions temporarily inflate the number of forwards per delivered message. As the buffer increases a

little, more copies get injected and more partial transfers occur before delivery succeeds, which raises overhead.

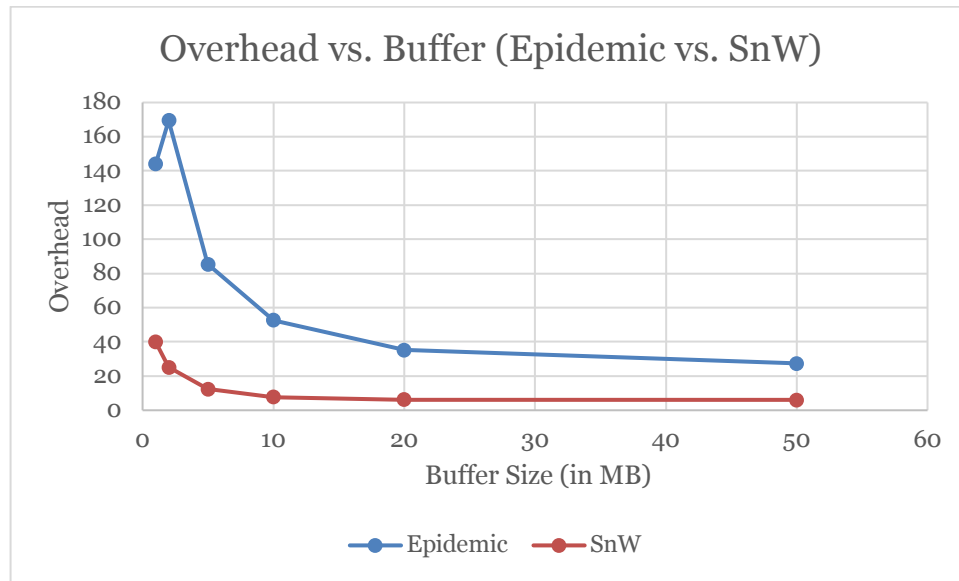


Figure 4: Forwarding overhead vs. buffer size (Epidemic vs. Spray-and-Wait).

3.1.4 Aborted Transfers and Hop Count

Aborted transfers indicate contacts that start but cannot finish due to short range/duration or contention. Spray and Wait consistently record fewer aborted transfers (e.g., at 1 MB, 5180 vs 9365 for Epidemic; at 20 MB, 12750 vs 22858), consistent with its capped replication and lower channel contention. Average hop count trends lower for Spray and Wait (~ 2.1 – 2.13 by 20–50 MB) than for Epidemic (declining from 4.58 at 1 MB to 2.26 at 50 MB).

3.2 Analysis of energy-aware protocols

3.2.1 Delivery Rate vs. Scan Interval

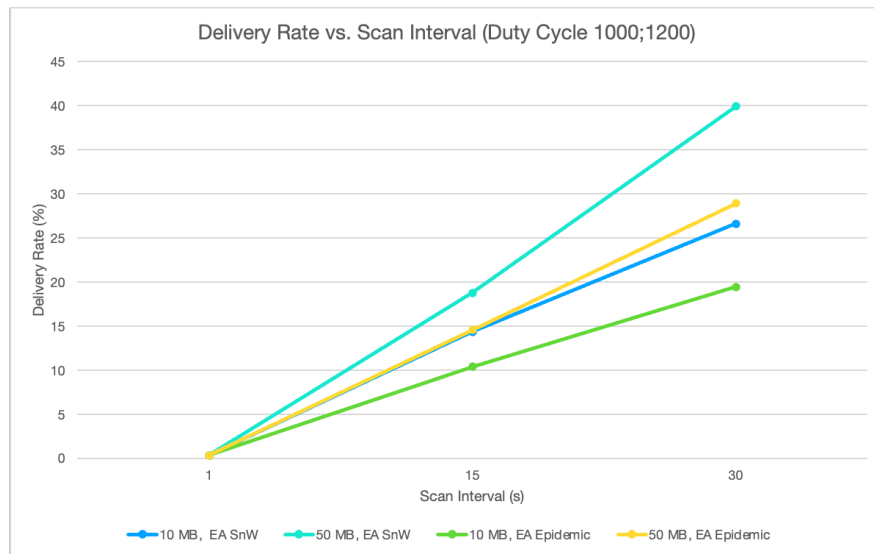


Figure 5: Delivery Rate vs. Scan Interval (Duty Cycle 1000;1200)

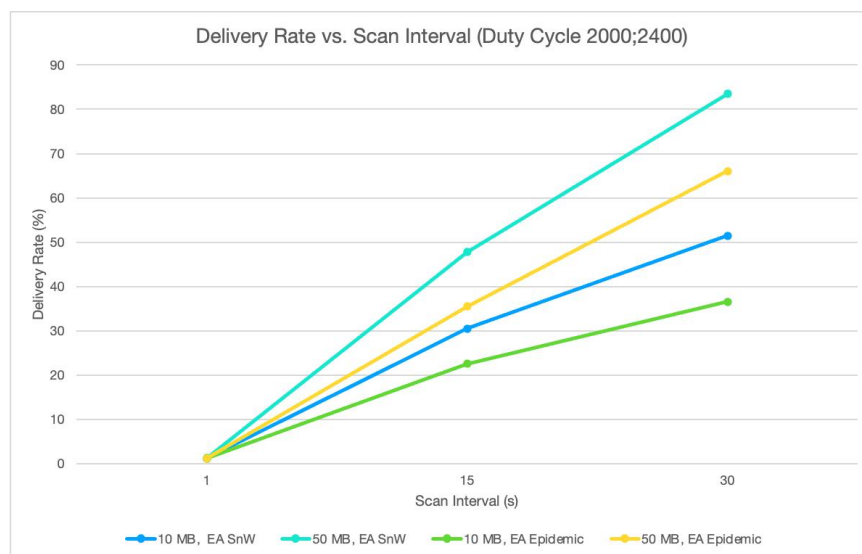


Figure 6: Delivery Rate vs. Scan Interval (Duty Cycle 2000;2400)

When the scan is set to 1s, delivery is near zero across all configurations (about 0–1%) since many short contacts are missed or radios don't stay awake long enough to finish a transfer. Increasing scanning to 15 s and 30 s causes a large jump in delivery, and at that point, buffer size starts to matter.

We can observe that 50 MB consistently outperforms 10 MB since nodes can carry more messages between brief encounters. Using a duty cycle with a longer awake period, 2000;2400 ms (i.e., longer continuous ON windows) shifts the entire curve upward, as shown in

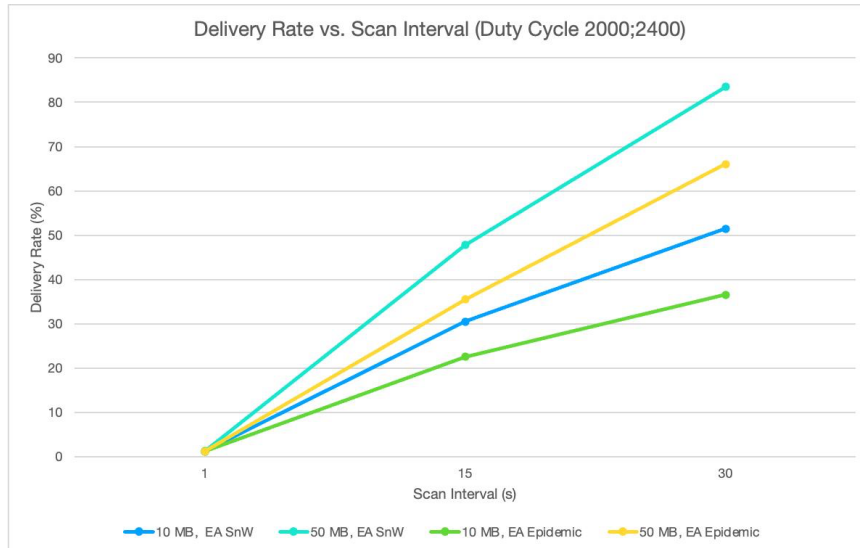


Figure 6. At a 30s scan interval, a 50 MB buffer reaches about 83.5% delivery rate with Spray-and-Wait. For duty cycle of 1000;1200 ms, the delivery rate is much lower. EA Spray-and-Wait achieves higher delivery with less forwarding compared to EA Epidemic.

3.2.2 Median Delay vs. Scan Interval

When we scan from 1 → 15 → 30 s, more multi-hop paths are discovered and actually finish. Consequently, the typical delivery time goes up. Once scanning is frequent enough, a 50 MB buffer tends to have a higher delay than 10 MB, since more messages are kept and some take longer routes before they finally arrive. For the same settings, EA Spray-and-Wait usually has equal or lower median delay than EA Epidemic.

3.2.3 Forward Overhead vs. Scan Interval

At scan = 1s, almost nothing is delivered, so the overhead is very high. As we move to 15s and 30s, delivery rises and the overhead drops. Across all duty cycles and buffers, EA Spray-and-Wait uses fewer extra forwards than EA Epidemic.

4 Discussion

In our city-scale DTN simulation, SnW consistently outperformed Epidemic across all buffer sizes: it achieved higher delivery, lower delay, and far lower forwarding overhead. Delivery improves rapidly as the buffer grows, with a clear knee around 10–20 MB. Beyond this, gains are small while delay keeps increasing. Practically, Spray and Wait with 10–20 MB per node offers the best delivery, delay and efficiency trade-off. Epidemic needs much larger buffers to approach Spray and Wait’s delivery and remains costly in terms of buffer and transmission overhead.

When introducing **energy-awareness** through duty cycles and scan intervals, the trade-offs became even more apparent. Both protocols suffer when radios scan too frequently (e.g., every 1s), as nodes fail to complete transfers before going back to sleep. Increasing the scan interval to 15–30s significantly improves delivery, especially when combined with larger buffer sizes that allow nodes to retain messages during missed opportunities.

Between the two, Spray-and-Wait maintained its advantage: it achieved higher delivery rates at equivalent energy settings, while also keeping forwarding overhead substantially lower. For instance, under the more energy-permissive duty cycle (2000;2400 ms) and larger buffer (50 MB), Spray-and-Wait reached 83.5% delivery with much lower overhead compared to Epidemic. Epidemic's performance did improve under relaxed energy constraints, but its energy cost remained disproportionately high. This supports our hypothesis that **Spray-and-Wait is more resource-efficient in constrained environments**, while Epidemic becomes competitive only when both buffer and energy are abundant.

In terms of **reliability and generalisability**, our results align with previous studies that highlighted Spray and Wait's efficiency advantage under constraints and Epidemic's strength only under unconstrained scenarios. However, it is important to note that our simulations were limited to a single urban mobility model (Helsinki map) and synthetic traffic generation. Different mobility patterns, traffic loads, or heterogeneous node energy capacities could somewhat shift the balance.

Future work should include more extensive testing, particularly of variable values (buffer sizes, scan intervals, and duty cycles) would provide a more complete view of protocol behaviour. Our current study used a limited set of values, and expanding this range could uncover non-linear effects or performance thresholds that were not visible here. It would also be in our interest to explore other routing protocols, such as PROPHET, with the same constraints.

In summary, Spray and Wait is more suitable for resource-constrained DTN's when compared to Epidemic, particularly in vehicular DTNs. Furthermore, energy-aware experiments highlight that adaptive duty cycles and scan intervals are key aspects when it comes to balancing performance and energy consumption, which leads us to consider not only routing logic but also efficient resource management.

References

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Appendix

We used ChatGPT to improve the language of the sections: abstract, introduction and discussion. We used the following prompt:

“Background: I am a student in a course entitled “Research Methodology and Scientific Writing” and I am using this tool for a project. My teacher allows me to use AI tools to

improve language aspects of my writing, but not to generate new ideas.

Instruction: Please suggest how to enhance the “input text” below, pointing out any errors, and provide feedback on the clarity, spelling, sentence structure, and grammar. If appropriate, list up to three spelling errors. If appropriate, list and explain up to three grammatical errors. If appropriate, list up to three suggestions for a change in style, coherence, or logical flow. Provide a suggestion for rephrased text.

Specific: Focus on refining my existing ideas. Ensure no ideas are removed or lost. Ensure that the content remains unchanged and no new concepts are introduced.

Input text: [insert paragraph text here]”